

STATISTICAL INFERENCE GUIDE

Confidence intervals

Prepare. Determine the *point estimate* and the appropriate distribution. Identify the *confidence level* and the corresponding z^* or t^* .

Check. Verify that the *conditions for inference* are satisfied.

Calculate. Calculate the *standard error* and *margin of error*. Construct the confidence interval.

Conclude. Interpret the interval in context.

Hypothesis testing

Prepare. State the *null and alternative hypotheses*. Make note of the *null value*. Identify the appropriate *test statistic* and its *distribution*. State the *significance level* (α).

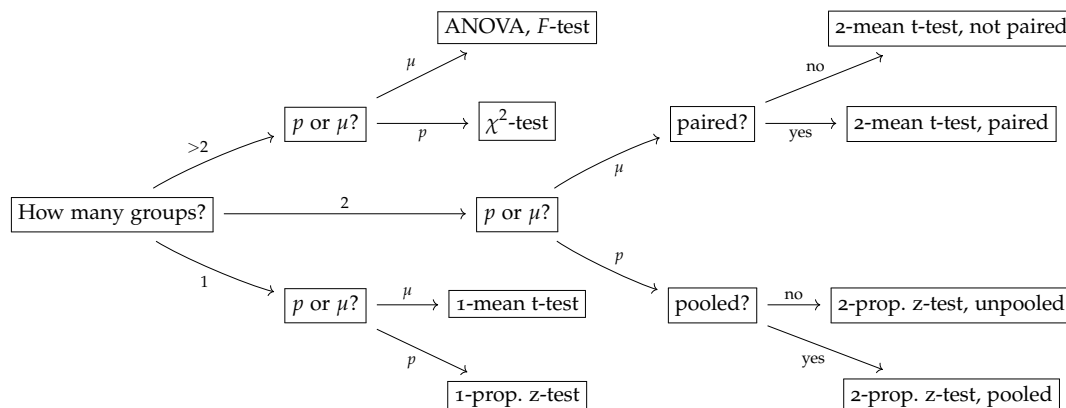
Check. Verify that the *conditions for inference* are satisfied.

Calculate. Calculate the *standard error*, *test statistic*, and *p-value*.

Conclude. If the p-value is less than α , *reject the null hypothesis* and interpret the result in context.

If the p-value is greater than or equal to α , *fail to reject the null hypothesis* and interpret the result in context.

Inference flowchart



Conditions for inference

Statistical inference generally requires *independent observations* and a *large enough sample size*.

Proportions: $np \geq 10$ and $n(1-p) \geq 10$.

Means: $n \geq 30$ with no extreme outliers, or population is nearly normal.

Confidence and significance

$$\boxed{\text{confidence}} = 1 - \boxed{\text{significance}}$$

90% confidence: $z^* = 1.65$, $\alpha = 0.10$

95% confidence: $z^* = 1.96$, $\alpha = 0.05$

98% confidence: $z^* = 2.33$, $\alpha = 0.02$

99% confidence: $z^* = 2.58$, $\alpha = 0.01$

Confidence intervals

$$\boxed{\text{point estimate}} \pm \underbrace{\boxed{\text{critical value}} \times SE}_{\text{margin of error}}$$

Test statistics

$$Z \text{ or } T = \frac{\boxed{\text{point estimate}} - \boxed{\text{null value}}}{SE}$$

$$\chi^2 = \sum_i \frac{(Ob. - Ex.)^2}{Ex.}$$

$$F = \frac{MSG}{MSE}$$

Normal or t-distribution?

A normal distribution may be used in place of a *t*-distribution provided (1) the degrees of freedom are at least 30, or (2) the *population's standard deviation* (σ) is known.

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Common formulas

Procedure	Parameter	Point estimate	Standard error
1-prop. C.I.	p	$\hat{p}$	$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
1-prop. z-test, (null value is p_0)	p	$\hat{p}$	$\sqrt{\frac{p_0(1-p_0)}{n}}$
2-prop. C.I. & 2-prop. z-test, unpooled	$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$	$\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
2-prop. z-test, pooled (null value is 0)	$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$	$\sqrt{\frac{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})}{n_1} + \frac{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})}{n_2}}$
1-mean C.I. & 1-mean t -test	μ	$\bar{x}$	$\frac{s}{\sqrt{n}}$
2-mean C.I., paired & 2-mean t -test, paired	μ_{diff}	$\bar{x}_{\text{diff}}$	$\frac{s_{\text{diff}}}{\sqrt{n_{\text{diff}}}}$
2-mean C.I., not paired & 2-mean t -test, not paired	$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	$\sqrt{\frac{(s_1)^2}{n_1} + \frac{(s_2)^2}{n_2}}$

Distributions

Test statistic	Distribution	Degrees of freedom
Z-score (proportions)	normal distribution	
T-score (means)	t -distribution	$n - 1$
2-sample T-score (means)	t -distribution	$\min\{n_1 - 1, n_2 - 1\}$
1-way χ^2 (proportions)	χ^2 -distribution	$\boxed{\text{\# of levels}} - 1$
2-way χ^2 (proportions)	χ^2 -distribution	$\boxed{\text{df for variable 1}} \times \boxed{\text{df for variable 2}}$
F (means)	F -distribution	$\text{df}_1 = \boxed{\text{\# of groups}} - 1, \text{df}_2 = n - \boxed{\text{\# of groups}}$